

## JEFF WIFI VOUCHER MANAGER

Version: v1\_STABLE

Project Documentation

Purpose:

JeffWiFi Voucher Manager is a Python-based voucher management system designed for a MikroTik hotspot business.

Project capabilities:

- Generate WiFi vouchers
- Support multiple plans
- Create MikroTik import files
- Maintain master inventory
- Track voucher lifecycle
- Track generation batches
- Protect voucher status transitions
- Run inventory integrity checks
- Display inventory dashboard

Folder Structure:

JeffWiFi-Voucher/

main.py

config.py

generator.py

inventory.py

exporter.py

voucher\_manager.py

inventory\_dashboard.py

imports/

Contains MikroTik RouterOS import files.

vouchers/

Contains customer voucher lists.

inventory/

Contains:

- master\_inventory.csv
- batch\_history.csv

cards/

Reserved for future printable voucher cards.

backups/

Reserved for future automated backups.

Python Files:

config.py:

Central configuration file. Contains project paths and voucher plan definitions.

generator.py:

Creates random voucher codes according to plan prefixes.

inventory.py:

Creates and updates the master inventory CSV file.

exporter.py:

Creates MikroTik RouterOS .rsc import files.

main.py:

Main application workflow:

1. Select plan
2. Select quantity
3. Generate vouchers
4. Save voucher list
5. Update inventory
6. Create MikroTik import file
7. Record batch history

voucher\_manager.py:

Manages voucher lifecycle:

AVAILABLE -> SOLD -> USED

Rules:

AVAILABLE vouchers can be sold.

Only SOLD vouchers can become USED.

USED vouchers cannot be sold again.

inventory\_dashboard.py:

Displays:

- Total vouchers
- Status summary
- Plan summary
- Last generated batch
- Data integrity check

Data Integrity Checks:

The system verifies:

- Duplicate voucher codes
- Invalid status transitions
- Missing required dates

Voucher Lifecycle:

AVAILABLE

|

v

SOLD

|

v

USED

Current Version:

v1\_STABLE

Future improvements:

- Revenue tracking
- Pricing
- GUI interface
- Automatic MikroTik API synchronization
- Printable voucher cards
- Cloud database support

Development Log:

This file should be updated after every major change.